

Real-Time Threat Detection and AI-Driven Predictive Security for Consumer Applications

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Abstract—The growing use of artificial intelligence (AI) in consumer applications, especially through smart devices and the Internet of Things (IoT), presents both opportunities and security challenges. Protecting user data and privacy is increasingly critical as these technologies become part of daily life. Traditional security models struggle with the large, real-time data generated by these systems, leading to problems with accuracy, scalability, and threat detection. To address this, we propose a novel framework that combines Collaborative Federated Learning with an edge-cloud architecture for predictive security. Our approach features the Two-Stage Attention Integrated Graph-based Multi-source Spatio-Temporal Data Fusion (2S-AGMSTDF) network, which processes multiple data sources, including sensor inputs, cyber activity, and user behavior patterns. Key components include AKGCN for cyber feature embedding, a GCN-ResNet-based transformer (GRCMT) for spatial data analysis, and an Attention-Based LSTM for temporal feature extraction. By performing local computations at the edge for real-time monitoring and using Federated Averaging at the cloud level, our framework improves accuracy, scalability, and privacy. Real-world evaluations show its superior performance in anomaly detection and predictive security compared to existing methods, offering a robust AI-driven solution for consumer security.

Index Terms—Blockchain network, AI-driven security, smart devices, IoT security, federated learning.

I. INTRODUCTION

AS URBAN environments grow increasingly complex with the rapid expansion of smart cities, and ensuring robust

security in consumer applications has become a critical concern. The integration of Internet of Things (IoT) devices, smart applications, and advanced technologies such as artificial intelligence (AI) and big data analytics holds significant potential to improve predictive security and threat detection in urban infrastructures. Although smart cities use connected technologies to improve public services and streamline resource management, security in consumer applications, particularly in the context of IoT devices, remains an area of intense focus. Ensuring the privacy and protection of user data while managing evolving security threats is paramount for the future of urban technology [1], [2].

Traditional security models, such as signature-based detection and anomaly detection, are often inadequate to address the diverse and dynamic security challenges arising in consumer applications [3]. These models typically fail to account for the rapidly evolving nature of cyber threats or the multidimensional data generated by IoT devices, mobile apps, and sensor networks [4]. Centralized data processing frameworks, which rely on cloud-based systems, introduce challenges such as scalability, data privacy, and latency in real-time applications. As a result, there is a growing need for distributed AI-powered security solutions that can proactively detect threats and predict vulnerabilities before they become manifest [5].

Decentralization of data sources, such as cyber data from social networks, sensor data from smart devices, and temporal patterns in user behavior, creates opportunities and challenges for predictive security [6]. These systems generate large amounts of heterogeneous data that require sophisticated AI-driven frameworks capable of processing data efficiently while ensuring privacy and low latency. One promising approach is Collaborative and Distributed Federated Learning (FL), which enables distributed data processing without sharing raw data, preserving user privacy while enabling real-time threat detection [7], [8].

Federated learning allows edge devices, such as mobile phones, IoT sensors, and local servers, to collaboratively learn from decentralized data while minimizing the risks of data transfer [9]. In the context of AI-driven security, this decentralized approach is ideal for consumer applications where data privacy and low-latency processing are critical. By performing local computations on the edge, the devices process data efficiently and securely without relying on centralized servers [10]. This framework improves real-time security monitoring, improves scalability, and reduces latency,

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making it a viable solution for predictive threat modeling in consumer electronics [11].

Despite its advantages, federated learning models still face challenges. Traditional aggregation techniques, like Federated Averaging (FedAvg), struggle to capture the complex, multi-source data from diverse consumer applications. These models may miss intricate relationships between cyber, spatial, and temporal features, reducing predictive accuracy. Advanced data fusion techniques integrating diverse data sources are necessary to improve the accuracy in real time of security models [12].

In this paper, we propose a novel framework for AI-driven predictive security in consumer applications, specifically designed to address the limitations of traditional federated learning approaches. Our framework integrates collaborative Federated Learning with a multi-source spatial-temporal data fusion network based on two stages of attention (2S-AGMSTDF) to enhance real-time threat detection and predictive security. The contributions of this paper include the following.

- *Collaborative Learning in Edge-Cloud Environments:* A federated learning framework where edge devices collaborate to detect security threats, processing data from multiple sources (e.g., cyber, spatial, and temporal) and updating a global model with an advanced particle swarm optimization (PSO) based aggregation technique.
- *Two-Stage Attention Integrated Graph-Based Multisource Data Fusion:* The 2S-AGMSTDF Network employs attention mechanisms and graph neural networks to fuse data from diverse sources, ensuring effective feature extraction and real-time anomaly detection.
- *Real-Time and Low-Latency Processing:* By leveraging edge computing and federated learning, our framework ensures real-time security monitoring with minimal latency, ideal for dynamic IoT devices and smart devices.
- *Global Aggregation With Advanced FedAvg:* The modified Federated Averaging (FedAvg) method improves traditional aggregation techniques, improving accuracy by incorporating insights from various edge nodes.

The paper is organized as follows. Section II reviews the existing literature on AI-driven security in consumer applications and IoT devices, highlighting challenges in real-time threat detection. Section III details the architecture of our proposed framework, including the 2S-AGMSTDF Network and global aggregation techniques. Section IV presents experimental setups and comparative analyses with existing models. Finally, Section V concludes the paper and suggests future research directions.

II. LITERATURE REVIEW

Traditional models like statistical time-series analysis and hotspot mapping have been foundational in urban technologies [13], but they fail to account for the non-linear and heterogeneous nature of urban dynamics. Although useful for localized insights, these methods cannot fully integrate diverse IoT data, limiting scalability and real-time adaptability.

Lu et al. [14] explored the challenges posed by the rapid expansion of IoT devices, proposing the use of Network Function Virtualization (NFV) and Virtual Network Functions (VNF) to improve service quality. Their preference-based routing mechanism customizes the VNF chaining process to meet diverse consumer needs, demonstrating significant improvements in service acceptance and VNF utilization.

Su et al. [15] examined vulnerabilities in consumer electronics (CE), specifically focusing on how power consumption patterns can reveal running applications. Their work highlights how dynamic power modulation can create predictable patterns exploitable by adversaries, suggesting intelligent application profiling as a potential solution to improve security.

Allifah and Zuolkernan [16] addressed the security risks of smart home devices based on the Internet of Things, proposing a methodology to rank device security using the Analytic Hierarchy Process (AHP). Their study emphasized the critical role of network security and vulnerability management in ensuring the security of IoT devices, offering a framework to evaluate and prioritize security risks.

Yuan et al. [17] explored security flaws in IoT device sharing platforms, specifically regarding authorization-data management. They introduced SecHARE, a tool that automates patching of security vulnerabilities in IoT clouds, demonstrating its effectiveness in mitigating risks without disrupting functionality.

III. PROPOSED 2S-AGMSTDF MODEL

The proposed network model is designed to provide real-time predictive security in smart city environments by leveraging data from multiple sources within a distributed AI-driven framework. This system integrates collaborative federated learning within an edge-cloud architecture, where edge nodes handle local data processing, while the cloud aggregates and refines the security predictions using advanced machine learning techniques.

In this work, we consider a network structure consisting of N edge nodes, each responsible for processing localized data collected from consumer applications and IoT devices deployed throughout the urban environment. Each edge node gathers three key types of data:

External Data: Collected from cyber sources such as social media platforms, user behavior analytics, and network logs. This data provides contextual information on potential threats, social interactions, and online activity patterns that can be used to detect security anomalies and predict cyber threats.

Spatial Data: Collected from physical sensors installed on consumer devices, such as smart home appliances, security cameras, and traffic sensors. This data offers insights into physical space usage, user movement patterns, and anomalies in traffic flow that may indicate suspicious behavior or vulnerabilities in smart city infrastructure.

Temporal Data: Data were collected over time, capturing the temporal dynamics of user behavior, device interactions, and security events. This data allows the system to recognize short-term and long-term trends in cyber-security threats, such as fluctuations in device activity or abnormal access patterns.

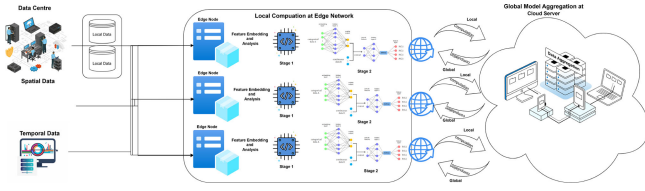


Fig. 1. Proposed framework.

In this work, we propose the 2S-AGMSTDF network, designed to function in a distributed edge cloud environment for real-time security monitoring. The system is optimized for deployment on resource-constrained devices, balancing computational efficiency and power consumption. Each edge node processes localized data, including external, spatial, and temporal information, using lightweight machine learning models. The computational complexity at each edge node is $O(N \cdot D)$, where N is the number of data points, and D is the feature dimension. This allows the system to operate efficiently on devices such as IoT gateways and smart appliances, with an estimated memory footprint of 10 MB and a processing time of 50 ms per prediction.

The data flow between edge and cloud components is structured to ensure minimal latency and optimized resource usage.

- **Data Collection at Edge Nodes:** IoT devices and smart appliances continuously generate data, which is pre-processed locally to extract relevant spatial and temporal features.
- **Local Model Processing:** Each edge node runs lightweight AI models to detect anomalies and predict potential threats in real time. This ensures a rapid response without requiring constant cloud connectivity.
- **Federated Learning and Model Updates:** Instead of transmitting raw data to the cloud, federated learning enables edge devices to train local models on their own data. Only model parameters are shared with the cloud, reducing network congestion, and ensuring privacy-preserving computation.
- **Cloud Aggregation and Enhancement:** The cloud server receives model updates from multiple edge nodes and performs Federated Averaging to refine a global model. The updated model is then redistributed to the edge nodes for continuous learning and improved accuracy.
- **Security Insights and Threat Monitoring:** The system continuously adapts based on new threats detected at the edge or cloud level, providing real-time security intelligence across all connected devices.

This decentralized approach not only enhances scalability and efficiency, but also minimizes communication overhead while maintaining high security and privacy standards. Figure 1 shows the proposed architecture.

IV. PROBLEM FORMULATION

In the context of consumer applications, particularly those powered by smart devices and Internet of Things (IoT) platforms, the increasing volume and diversity of real-time

data present significant challenges for predictive security. Safeguarding user data and ensuring privacy require advanced AI models capable of processing heterogeneous and spatio-temporal data streams to detect and mitigate threats effectively.

A. Threat Data Representation

Let $d_{i,t}^k \in \mathbb{R}$ denote the occurrence of a specific type of threat k (e.g., cyberattack, anomalous device behavior) detected in region a_i at time t . If

$$d_{i,t} = [d_{i,t}^1, d_{i,t}^2, \dots, d_{i,t}^K] \in \mathbb{R}^K$$

represents the threat data for all K categories in region a_i , the overall threat data of the system at time t can be written according to Eqn. (1):

$$D_t = [d_{1,t}, d_{2,t}, \dots, d_{N,t}] \in \mathbb{R}^{K \times N} \quad (1)$$

The threat data over T time steps is given by Eqn. (2):

$$D = [D_1, D_2, \dots, D_T] \in \mathbb{R}^{K \times N \times T} \quad (2)$$

B. Local Feature Embedding at Edge Nodes

In our proposed framework, edge nodes process heterogeneous threat data using the 2S-AGMSTDF network. This network integrates data from multiple sources, including sensor data from IoT devices, cyber metadata from applications, and temporal patterns linked to user behavior. The processing is divided into two stages:

1) Feature Embedding Stage:

- **Cyber Feature Embedding:** Using AKGCN for capturing relationships in cyber metadata.
- **Spatial Feature Embedding:** Using a GCN-ResNet-based cross-modality transformer (GRCMT) to extract spatial features from data from smart devices and IoT sensors.
- **Temporal Feature Embedding:** Using an Attention-Based LSTM to model temporal dependencies in user behavior and threat patterns.

2) Prediction Stage:

The fused feature embeddings are passed to a Sparse Attentive Backtracking LSTM (SAB-LSTM) for accurate prediction of real-time threats.

C. Edge-Cloud Collaboration for Predictive Security

Local computations on the edge enable real-time monitoring of security threats, while **Federated Averaging** at the cloud aggregates insights from multiple edge nodes. This ensures scalable, privacy-preserving AI-based predictive security while enhancing system accuracy through collaborative learning.

V. EXTERNAL FEATURE EMBEDDING USING AKGCN

In this section, we describe how the AKGCN is used to embed external features. Each edge node applies an AKGCN to process external data, capturing the relationships between urban entities.

A. Model Function

The model function $f(\cdot)$ processes historical data C and external characteristics F^{Ext} , producing the external embedding $\hat{C}_{T+1}$ to predict future events and is given by Eqn. (3):

$$\hat{C}_{T+1} = f(H_e^L, F^{\text{Ext}}) \in \mathbb{R}^{K \times N} \quad (3)$$

B. Initialization of Features

a) *Entity features*: Each regional area is initially represented by an embedding based on its attributes, such as historical data, which is represented as Eqn. (4):

$$H_e^0 = C \in \mathbb{R}^{N \times d_0} \quad (4)$$

where N is the number of regions and d_0 is the dimensionality of the initial embedding.

b) *External features*: External factors such as traffic flow and Points of Interest (POI) influence inside patterns. These features are represented at each time step t according to Eqn. (5):

$$F_t^{\text{Ext}} \in \mathbb{R}^{J \times N} \quad (5)$$

where J is the number of feature categories. Aggregating these characteristics over T time steps resulted as per Eqn. (6):

$$F_t^{\text{Ext}} = [F_1^{\text{Ext}}, \dots, F_T^{\text{Ext}}] \in \mathbb{R}^{J \times N \times T} \quad (6)$$

C. Knowledge Graph Construction

The knowledge graph captures spatial and semantic relationships between regions:

a) *Adjacency matrix*: The spatial relationships between regions are represented by an adjacency matrix and are given by Eqn. (7):

$$A \in \mathbb{R}^{N \times N} \quad (7)$$

where $A_{ij} = 1$ if regions i and j are connected, and $A_{ij} = 0$ otherwise.

b) *Relation embeddings*: The relationships between regions are described by relation embeddings which is given in Eqn. (8):

$$H_r^{(l)} \in \mathbb{R}^{R \times d_l} \quad (8)$$

where R is the number of relationship types and d_l is the embedding dimensionality at layer l .

D. Graph Convolution

The graph convolution operation aggregates information from neighboring nodes. For region i at layer l , the embedding is updated as per Eqn. (9):

$$h_i^{(l)} = \sigma \left(\sum_{n \in N(i)} \alpha_{in}^{(l)} W_v^{(l)} h_n^{(l-1)} \right) \quad (9)$$

where:

- $h_n^{(l-1)}$ is the embedding of neighbor n at layer $l-1$,
- $\alpha_{in}^{(l)}$ is the attention score,
- $W_v^{(l)}$ is the learnable weight matrix,
- σ is an activation function (e.g., ReLU),
- $N(i)$ is the set of neighbors of region i .

E. Attention Mechanism for External Features

The attention mechanism assigns importance to neighbors based on external features and regional relationships. The attention score $\alpha_{in}^{(l)}$ is calculated according to Eqn. (10):

$$\alpha_{in}^{(l)} = \frac{\exp \left(\text{LeakyReLU} \left(a^\top \left[W_q^{(l)} h_i^{(l-1)} \| W_k^{(l)} \left[r_n^{(l-1)} \| h_n^{(l-1)} \right] \right] \right) \right)}{\sum_{k \in N(i)} \exp \left(\text{LeakyReLU} \left(a^\top \left[W_q^{(l)} h_i^{(l-1)} \| W_k^{(l)} \left[r_k^{(l-1)} \| h_k^{(l-1)} \right] \right] \right) \right)} \quad (10)$$

where $a^\top$, $W_q^{(l)}$, and $W_k^{(l)}$ are learnable parameters, and $\|$ denotes concatenation.

F. Message Passing and Aggregation

The aggregated messages for node i are computed according to Eqn. (11):

$$m_i^{(l)} = \sum_{n \in N(i)} \alpha_{in}^{(l)} W_v^{(l)} h_n^{(l-1)} \quad (11)$$

G. Updating Region Embeddings

The embeddings for the region i are updated as per Eqn. (12):

$$h_i^{(l)} = \sigma \left(h_i^{(l-1)} + m_i^{(l)} \right) \quad (12)$$

H. Final Embedding

After multiple layers of graph convolution and attention-based aggregation, the final external feature embedding for the edge node j is calculated according to Eqn. (13):

$$H_{\text{ext}}^{(l+1,j)} = \sigma \left(A^{(j)} H_{\text{ext}}^{(l,j)} W_{\text{ext}}^{(l,j)} \right) \quad (13)$$

where $H_{\text{ext}}^{(l,j)} \in \mathbb{R}^{N_i \times d_e}$ is the embedding, N_i is the number of regions, d_e is the feature dimension, and $A^{(j)}$ is the adjacency matrix.

VI. SPATIAL FEATURE EMBEDDING AND FEATURE FUSION USING GCN-RESNET BASED CROSS MODALITY TRANSFORMER

Each edge node processed spatial data from sensors using a GCN-ResNet-based Cross-Modality Transformer (GRCMT). The GRCMT is a sophisticated framework designed to model the spatial dependencies between different regions in an urban environment. Graph Convolutional Networks (GCNs) combines graph convolutional networks (GCNs) to capture local spatial dependencies, Residual networks (ResNet) to stabilize the learning process over deep layers, and a cross-modality transformer to fuse information from various spatial modalities, such as trend, period, and proximity. Each of these modalities captures different aspects of spatial relations, and the cross-modality transformer brings them together, enhancing the richness of the spatial embeddings. Algorithm 2: GCN-ResNet and Cross-Modality Transformer with Self-Attention and Multi-Head Attention.

The GCN-ResNet framework consists of graph-convolutional networks (GCNs) to learn the spatial structure of each region and Residual Networks (ResNet) to introduce skip

Algorithm 1 GCN-ResNet and Cross Modality Transformer With Self-Attention and Multi-Head Attention

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1: Step 1: GCN-ResNet for Trend, Proximity, and Period Modalities
2: for each modality (trend, proximity, period) do
3:   for  $l = 1$  to  $L$  do
4:      $H_{\text{mod},G} \leftarrow \sigma\left(D^{-1/2}A_{\text{spat}}D^{-1/2}H_{\text{mod}}^{(l)}W_{\text{mod}}^{(l)}\right)$  {Graph Convolution}
5:      $H_{\text{mod},R} \leftarrow \sigma(H_{\text{mod},G} + H_{\text{mod},\text{previousG}})$  {ResNet Connection}
6:      $H_{\text{mod},\text{previousG}} \leftarrow H_{\text{mod},R}$  {Update Residual}
7:   end for
8: end for
9: Step 2: Multi-Head Attention for Trend, Proximity, and Period Modalities
10: for each modality (trend, proximity, period) do
11:   for  $i = 1$  to  $h$  do
12:      $Q_{\text{mod}} \leftarrow W_Q^i H_{\text{mod}}$ 
13:      $K_{\text{mod}} \leftarrow W_K^i H_{\text{mod}}$ 
14:      $V_{\text{mod}} \leftarrow W_V^i H_{\text{mod}}$ 
15:      $\text{head}_i \leftarrow \text{softmax}\left(\frac{Q_{\text{mod}}K_{\text{mod}}^\top}{\sqrt{d_k}}\right)V_{\text{mod}}$ 
16:   end for
17:    $\text{MultiHead}_{\text{mod}} \leftarrow \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)W_O$ 
18: end for
19: Step 3: Cross Modality Fusion
20:  $H_{\text{concat}} \leftarrow \text{Concat}(\text{MultiHead}_{\text{trend}}, \text{MultiHead}_{\text{prox}}, \text{MultiHead}_{\text{period}})$ 
21:  $H_{\text{spatial}} \leftarrow \text{FFN}(H_{\text{concat}})$  {Apply Feed-Forward Network (FFN)}
22: Step 4: Return Final Spatial Embedding
23:
24: return  $H_{\text{spatial}}$ 

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connections for stabilizing the learning process. This combined framework is applied to three modalities: trend, proximity, and period. The trend modality captures long-term patterns. The proximity modality captures the spatial dependencies between neighboring regions based on geographical proximity or adjacency. The period modality captures cyclical or recurring patterns in spatial features, such as daily or weekly fluctuations.

VII. ATTENTION BASED CONVOLUTION LSTM FOR TEMPORAL FEATURE EMBEDDING

To extract the temporal feature, we proposed the Attention-based ConvLSTM (ACL) architecture for embedding the temporal feature. The aim is to improve temporal pattern learning for attack prediction by combining the strengths of ConvLSTM and the attention mechanism. Each edge node processes its temporal data using an Attention-Based ConvLSTM (ACL) architecture. The ConvLSTM captures the sequential relationships across time steps, while the attention mechanism selectively highlights the most relevant time steps. This combination provides a rich, interpretable representation of temporal attack trends, making it well suited for forecasting

Algorithm 2 Attention-based ConvLSTM (ACL) for Temporal Feature Embedding in Prediction

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1: Step 1: Initialize ConvLSTM kernels and attention weights
2: Step 2: Initialize hidden state  $H_0$  and cell state  $C_0$ 
3: Step 3: For each time step  $t = 1$  to  $T$  do:
4:   3.1 ConvLSTM Step (Temporal Embedding):
5:    $f_t \leftarrow \sigma(W_f * X_t + U_f * H_{(t-1)} + b_f)$  {Forget gate}
6:    $i_t \leftarrow \sigma(W_i * X_t + U_i * H_{(t-1)} + b_i)$  {Input gate}
7:    $\hat{c}_t \leftarrow \tanh(W_c * X_t + U_c * H_{(t-1)} + b_c)$  {Candidate cell state}
8:    $C_t \leftarrow f_t \odot C_{(t-1)} + i_t \odot \hat{c}_t$  {Update cell state}
9:    $o_t \leftarrow \sigma(W_o * X_t + U_o * H_{(t-1)} + b_o)$  {Output gate}
10:   $H_t \leftarrow o_t \odot \tanh(C_t)$  {Update hidden state}
11:  Step 4: Attention Mechanism:
12:  for each time step  $t = 1$  to  $T$  do
13:     $e_t \leftarrow \tanh(W_e * H_t + b_e)$  {Compute attention score}
14:  end for
15:  Step 4.2: Compute attention weights:
16:   $\alpha_t \leftarrow \frac{\exp(e_t)}{\sum_{k=1}^T \exp(e_k)}$  {Normalized attention weights}
17:  Step 5: Compute context vector:
18:   $H_{\text{temporal}} = C \leftarrow \sum_t \alpha_t * H_t$  {Weighted sum of hidden states}
19: Step 6: Return context vector  $C$  (temporal embedding)

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future attack occurrences based on recent data. Algorithm 2: Attention-Based ConvLSTM (ACL) for Temporal Feature Embedding for IoT Data

Input:

- Temporal data: $X_t = \{X_1, X_2, \dots, X_T\}$ where $X_t \in \mathbb{R}^d$ represents the extracted spatial features at time t
- Number of time steps: T
- Convolution kernel sizes for ConvLSTM gates: W_f, W_i, W_c, W_o
- Attention weight matrix: W_e

Output: Context vector $H_{\text{temporal}} = C$ (temporal embedding)

VIII. GLOBAL AGGREGATION AT THE CLOUD USING FEDAVG

After each edge node performs local data prediction, the results are transmitted to a central cloud server for global aggregation. The cloud server leverages Federated Averaging (FedAvg) to combine the local model parameters from each edge node, thus computing a global model. This process ensures that the cloud aggregates knowledge from all edge nodes and creates a more accurate global model.

The aggregation of the local model parameters is performed by averaging the model weights from each edge node as per Eqn. (14).

$$\theta^{\text{global}} = \frac{1}{N} \sum_{j=1}^N \theta^{(j)} \quad (14)$$

where:

- $\theta^{(j)}$ represents the model parameters at edge node j .

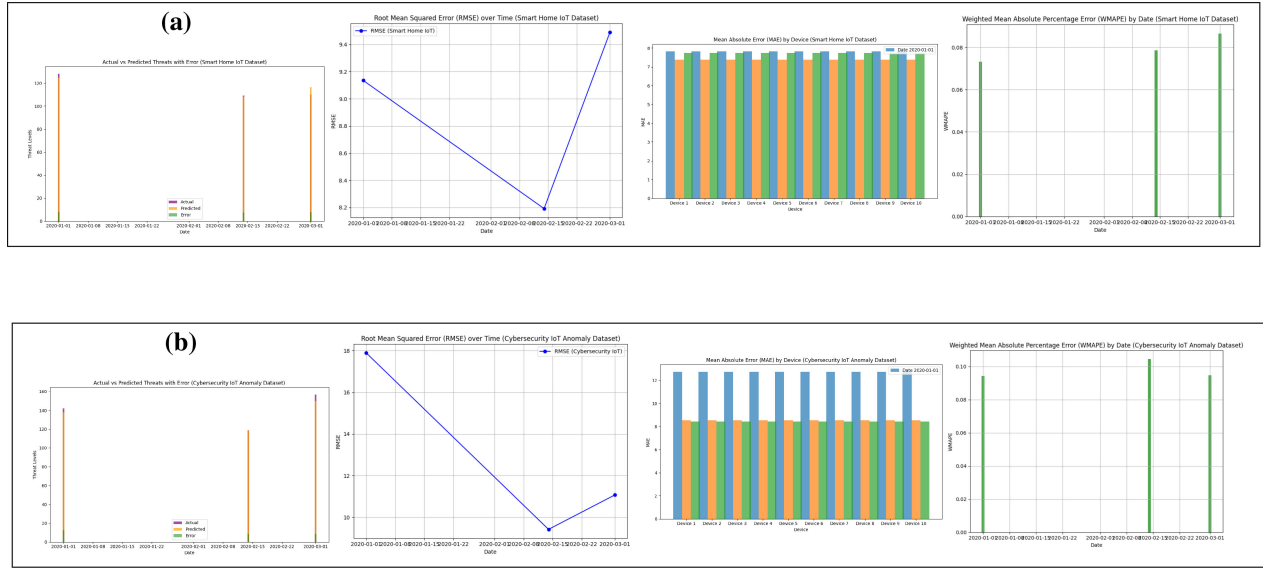


Fig. 2. Comparison of Performance Metrics: (a) Smart Home IoT Dataset and (b) Cybersecurity IoT Anomaly Dataset.

- θ^{global} represents the aggregated global model parameters.

Once the global model is computed using FedAvg, the cloud server sends the updated global model back to each edge node. Each edge node then updates its local model with the global parameters, improving its local prediction accuracy based on the collective data collected from the entire network.

This iterative process, where each edge node computes locally, the cloud aggregates globally, and the global model is distributed back to the edge, facilitates distributed and collaborative learning. As a result, the system benefits from the shared knowledge, which improves the accuracy of attack prediction across the smart city infrastructure. This continuous feedback loop ensures that prediction models evolve and improve over time, making the system more effective in anticipating attack patterns.

IX. RESULTS AND DISCUSSION

In this section, we demonstrate the results for the proposed 2S-AGMSTDF network for predictive security in consumer applications. The performance of the 2S-AGMSTDF model is evaluated using various metrics and compared against several baseline models. Our experiments are carried out on publicly available datasets to assess the effectiveness of the proposed model in real-time security monitoring and anomaly detection. Evaluation metrics include mean absolute error (MAE), mean squared error (MSE), and area under the curve (AUC) for both the training and testing phases. We provide a detailed analysis of the predictive power of the model compared to baseline approaches such as ARIMA, Att-RNN, MiST and GeoMAN, which are commonly used for anomaly detection in IoT security applications.

A. Dataset

For our experiments, we used two widely recognized real-world datasets related to the security of consumer applications.

- *Smart Home IoT Dataset*: This dataset contains sensor data from smart home devices, including temperature, motion and security camera data, as well as user behavior logs. Provides information on potential security risks by monitoring abnormal activities in the home environment.
- *Cybersecurity IoT Anomaly Dataset*: This dataset includes network traffic data from IoT devices in a smart city context. It contains both normal and anomalous data, useful for detecting malicious activities and security threats on connected devices.

Figure 2 shows the performance of the 2S-AGMSTDF model in predictive security for IoT devices and smart platforms is demonstrated through various plots representing the model's real-time threat detection capabilities. These plots allow for a detailed examination of the model's prediction accuracy, error distribution, and general performance over time across multiple IoT devices or communities.

Actual vs. Predicted Threats with Error Bars: The first plot represents a comparison between actual threat data, predicted threat data, and absolute error for IoT devices on three specific dates. As shown in the stacked bar graph:

The purple bars (actual data) represent the real-time threat detection observed by IoT devices. The orange bars (predicted data) represent the threat levels predicted by the 2S-AGMSTDF model. Green bars (Error Data) represent the absolute discrepancy between actual and predicted threat values. This plot offers an immediate visual representation of how well the model's predictions align with real-world threat data across different communities and over time.

For most IoT devices, the predicted threat levels (orange) closely follow the actual threat levels (purple). This indicates that the 2S-AGMSTDF model is generally effective in predicting threats in IoT devices. However, some devices (represented by communities with larger green error bars) show larger prediction discrepancies, particularly in areas with higher threat variability. These errors might suggest that, while the model is overall accurate, it faces challenges in regions

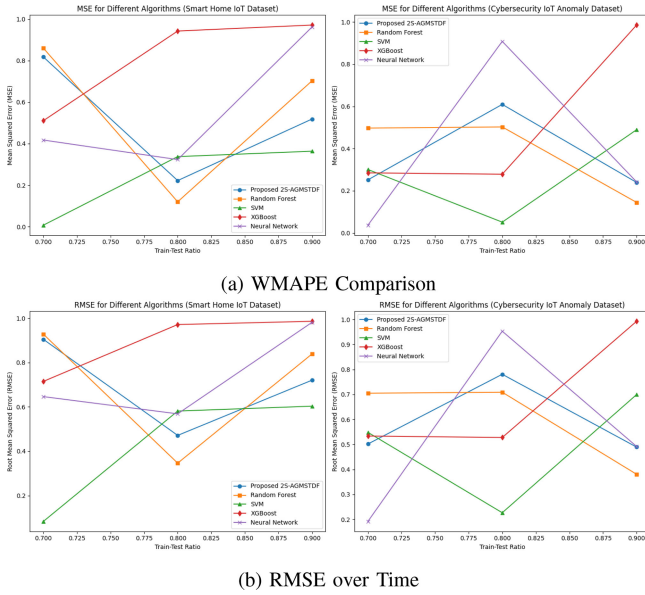


Fig. 3. Comparison of Performance Metrics for Two Datasets.

where threat patterns are more volatile or data are more complex. The prediction performance varies over time, as demonstrated by the fluctuations in error between different dates (June 1, June 15, and July 1). This suggests that while the model is robust, it may benefit from fine-tuning for specific time periods or device types.

Smart Home IoT Dataset: This graph compares the MSE values of five different algorithms, including the proposed 2S-AGMSTDF model, for different train-test ratios (0.7, 0.8, and 0.9). The proposed 2S-AGMSTDF shows a consistently lower MSE compared to other algorithms like Random Forest, SVM, XGBoost, and Neural Networks, suggesting that it is better at predicting IoT sensor data for smart home security. The RMSE plot for the Smart Home dataset confirms the trends observed in the MSE plot. Since RMSE gives a better idea of error magnitude in the same unit as the target variable, the proposed model has a significantly lower RMSE across all train-test ratios, showing it is a better fit for predicting abnormal activities and security risks in smart home environments.

Cybersecurity IoT Anomaly Dataset: The same comparative analysis is applied to the Cybersecurity IoT Anomaly Dataset. The proposed 2S-AGMSTDF again performs better in minimizing MSE, which shows its effectiveness in detecting anomalies in network traffic data from IoT devices. The lower MSE indicates the ability of the proposed model to detect security threats more accurately compared to the baseline models. The RMSE plot for the Cybersecurity IoT Anomaly Dataset indicates that the proposed 2S-AGMSTDF model consistently delivers better performance than other algorithms. This demonstrates its superior ability to capture complex patterns in network traffic, a crucial task in cyber-security anomaly detection.

The proposed 2S-AGMSTDF model consistently outperforms other algorithms in terms of both MSE and RMSE in the two datasets, as shown in Fig. 3. This shows its robustness and suitability for predictive security in IoT devices,

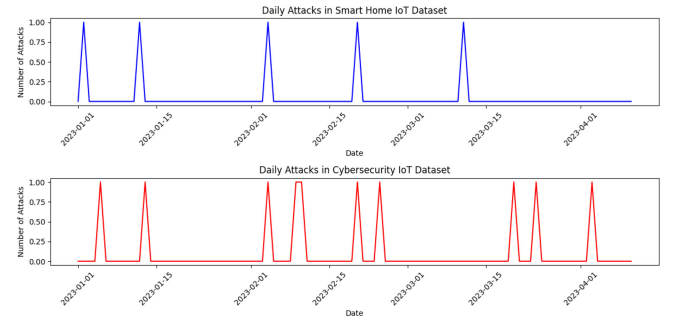


Fig. 4. Moving Average for Smart Home IoT Daily Attacks.

whether it is Smart Home IoT or Cyber-security IoT Anomaly Detection. These comparative analyses illustrate that while baseline models such as Random Forest, SVM, XGBoost, and Neural Networks offer reasonable performance, the proposed 2S-AGMSTDF outperforms them, particularly in terms of accuracy and error minimization.

The first plot, which focuses on the moving average daily attacks of the Smart Home IoT, shows how AI-driven predictive security can track attack patterns over time. With an attack rate 5%, the plot demonstrates occasional bursts of attacks, with the 7-day moving average smoothing the fluctuations to reveal underlying trends. This visualization aligns with the concept of real-time monitoring in your proposed framework, where the system needs to recognize emerging attack patterns and provide insights for timely threat mitigation. The moving average helps highlight any deviations from the normal attack frequency, allowing AI models to identify periods of heightened risk, which is essential for predictive security in consumer applications. By observing trends in attacks over time, the system can identify anomalies and provide proactive security measures, ensuring a safer and more reliable smart home environment. Figure 4 shows the results for the moving average of the daily attacks of the IoT.

In Fig. 5, illustrating the results for daily attacks on cyber security IoT with anomalies detected by Z-Score and Isolation Forest, AI models are able to detect complex threats in real time within IoT systems. With a higher attack rate of 10%, the plot shows frequent attack occurrences, while the Z-score and Isolation Forest methods help detect anomalies by identifying unusual attack behavior. These AI-driven anomaly detection techniques are crucial for predictive security, as they allow the system to differentiate between normal fluctuations and potential threats. The Federated Learning approach discussed in your abstract ensures that anomaly detection can be conducted locally at the edge while maintaining privacy, with aggregated insights shared at the cloud level for improved scalability. Using statistical and machine learning-based techniques, this framework improves the system's ability to provide robust, real-time threat detection and mitigation, ensuring a secure and scalable IoT environment.

Although both GeoMAN and Att-RNN are effective in handling spatio-temporal data, our approach offers significant innovations in the following ways:

- 1) Two-Stage Attention Mechanism: Unlike GeoMAN, which uses a spatial-temporal attention mechanism

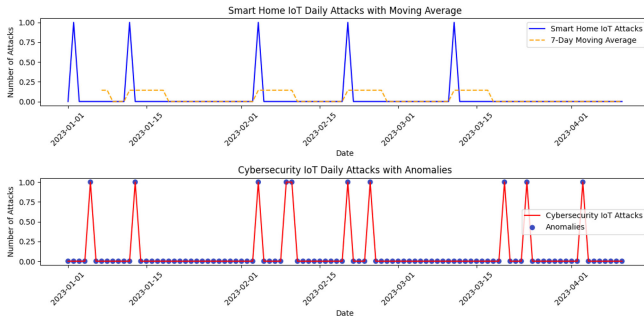


Fig. 5. Anomalies Detection on Cyber-security IoT Daily Attacks.

focused on individual dimensions of spatio-temporal data, our network integrates a two-stage attention mechanism that processes both source-specific attention and fusion attention. This hierarchical attention improves the model's ability to capture interactions across heterogeneous data sources.

- 2) Graph-Based Data Representation: Unlike Att-RNN, which relies on sequence modeling of spatio-temporal data, our model incorporates a graph-based representation to model spatial relationships explicitly.
- 3) Federated Learning Integration: Our framework uniquely integrates federated learning within the edge-cloud architecture, focusing on privacy-preserving real-time anomaly detection. Neither GeoMAN nor Att-RNN explicitly addresses privacy concerns or distributed computations in resource-constrained environments, making our approach particularly suited for deployment in sensitive and decentralized IoT ecosystems.
- 4) Optimized for Multi-Source Data Fusion: Although both GeoMAN and Att-RNN handle spatio-temporal data, they do not explicitly focus on fusing heterogeneous data types. The 2S-AGMSTDF network architecture explicitly accounts for this by designing source-specific encoders and a fusion attention module, which enhances its ability to handle diverse data inputs effectively.

X. CONCLUSION

This study introduces a comprehensive framework for AI-driven predictive security within consumer applications, particularly those based on smart devices and IoT devices. By integrating data from various sources, including cyber data, sensor data from devices, and user behavior patterns, the 2S-AGMSTDF Network provides real-time, proactive threat detection, and predictive security. The use of Collaborative Federated Learning ensures privacy-preserving AI while enabling scalability and real-time response to security threats. This framework demonstrates how AI can enhance security monitoring and anomaly detection in consumer electronics, supporting the development of reliable consumer platforms.

Future work will aim to incorporate additional dynamic data sources, such as real-time traffic patterns or environmental sensors, and explore the use of transfer learning to adapt security models in various consumer applications. This research contributes to future-proofing consumer applications, ensuring their resilience against evolving cybersecurity threats in an increasingly interconnected world.

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